

Below Sql DB and table setup with queries, joins, Views and CTEs using SSMS

laptop-ekoigt17 (SQL Server 16.0.100)

Databases

System Databases

Database Snapshots

CybersecurityDB

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Departments

Columns

DepartmentID (PK, I)

DepartmentName ()

Keys

Constraints

Triggers

Indexes

Statistics

dbo.Employees

dbo.Logs

Dropped Ledger Tables

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Security

Server Objects

Replication

Always On High Availability

Management

Integration Services Catalogs

```
USE CybersecurityDB;
GO

-- Step 2: Create Tables
-- Create Employees Table
CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY IDENTITY(1,1),
    FirstName NVARCHAR(50),
    LastName NVARCHAR(50),
    DepartmentID INT,
    Email NVARCHAR(100),
    IsActive BIT
);
GO

-- Create Departments Table
CREATE TABLE Departments (
    DepartmentID INT PRIMARY KEY IDENTITY(1,1),
    DepartmentName NVARCHAR(100)
);
GO
```

100 %

Results	Messages		
LogID	EmployeeID	LogTime	Activity
1	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
2	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
3	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
4	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3
5	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
6	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
7	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
8	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3

(No column name)

1

8

(No column name)

1

12

(No column name)

1

8

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8 ... | CybersecurityDB | 00:00:00 | 31 rows

Connect

laptop-ekoigt17 (SQL Server 16.0.100)

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Integration Services Catalogs

```
-- Create Departments Table
CREATE TABLE Departments (
    DepartmentID INT PRIMARY KEY IDENTITY(1,1),
    DepartmentName NVARCHAR(100)
);
GO

-- Create Logs Table (Security Logs for Cybersecurity)
CREATE TABLE Logs (
    LogID INT PRIMARY KEY IDENTITY(1,1),
    EmployeeID INT,
    LogTime DATETIME,
    Activity NVARCHAR(255),
    FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)
);
GO

-- Step 3: Insert Sample Data
-- Insert sample departments
INSERT INTO Departments (DepartmentName)
VALUES ('IT'), ('HR'), ('Finance');
GO
```

100 %

Results	Messages		
LogID	EmployeeID	LogTime	Activity
1	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
2	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
3	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
4	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3
5	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
6	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
7	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
8	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3

(No column name)

1

8

(No column name)

1

12

(No column name)

1

8

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8 ... | CybersecurityDB | 00:00:00 | 31 rows

Connect ▼

laptop-ekoigt17 (SQL Server 16.0.100)

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```
-- Insert sample employees
INSERT INTO Employees (FirstName, LastName, DepartmentID, Email, IsActive)
VALUES
('John', 'Doe', 1, 'johndoe@example.com', 1),
('Jane', 'Smith', 2, 'janesmith@example.com', 1),
('Sam', 'Brown', 3, 'sambrown@example.com', 0),
('Alice', 'Davis', 1, 'alicedavis@example.com', 1);
GO

-- Insert sample logs
INSERT INTO Logs (EmployeeID, LogTime, Activity)
VALUES
(1, '2025-03-10 08:30:00', 'Login from IP 192.168.1.1'),
(2, '2025-03-10 09:00:00', 'Login from IP 192.168.1.2'),
(1, '2025-03-11 10:15:00', 'File Accessed - sensitive data'),
(1, '2025-03-11 10:15:00', 'File Accessed - sensitive data'),
(3, '2025-03-12 12:00:00', 'Login from IP 192.168.1.3');
GO

SELECT * FROM Employees;
SELECT * FROM Departments;
SELECT * FROM Logs;

SELECT COUNT(*) FROM Employees;
```

100 %

Results Messages

LogID	EmployeeID	LogTime	Activity
1	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
2	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
3	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
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5	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
6	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
7	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
8	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3

(No column name)

1	8
---	---

(No column name)

1	12
---	----

(No column name)

1	8
---	---

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8 ... CybersecurityDB | 00:00:00 | 31 rows

Ready Ln 12 Col 22 Ch 22 INS

Connect ▼

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 - dbo.Departments
 - Columns
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 - DepartmentName (I)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.Employees
 - dbo.Logs
 - Dropped Ledger Tables
 - Views
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 - Synonyms
 - Programmability
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 - Always On High Availability
 - Management
 - Integration Services Catalogs

```
-- Insert sample logs
INSERT INTO Logs (EmployeeID, LogTime, Activity)
VALUES
(1, '2025-03-10 08:30:00', 'Login from IP 192.168.1.1'),
(2, '2025-03-10 09:00:00', 'Login from IP 192.168.1.2'),
(1, '2025-03-11 10:15:00', 'File Accessed - sensitive data'),
(3, '2025-03-12 12:00:00', 'Login from IP 192.168.1.3');
```

100 %

Results Messages

LogID	EmployeeID	LogTime	Activity
1	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
2	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
3	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
4	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3
5	1	2025-03-10 08:30:00.000	Login from IP 192.168.1.1
6	2	2025-03-10 09:00:00.000	Login from IP 192.168.1.2
7	1	2025-03-11 10:15:00.000	File Accessed - sensitive data
8	3	2025-03-12 12:00:00.000	Login from IP 192.168.1.3

(No column name)

1	8
---	---

(No column name)

1	12
---	----

(No column name)

1	8
---	---

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8 ... CybersecurityDB | 00:00:00 | 12 rows

Ready Ln 1 Col 1 INS

Connect - laptop-ekoigt17 (SQL Server 16.0.100)

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 - dbo.Departments
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 - DepartmentName (I)
 - Keys
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```

SELECT * FROM Logs;

SELECT COUNT(*) FROM Employees;
SELECT COUNT(*) FROM Departments;
SELECT COUNT(*) FROM Logs;

--Step 4: Queries
--Now, let's create some SQL queries.
--Get all employees with their department names:
SELECT e.EmployeeID, e.FirstName, e.LastName, d.DepartmentName, e.Email, e.IsActive
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID;
GO

SELECT e.FirstName, e.LastName, l.Activity, l.LogTime
FROM Employees e
JOIN Logs l ON e.EmployeeID = l.EmployeeID
ORDER BY l.LogTime DESC;
GO
  
```

100 %

Results Messages

EmployeeID	FirstName	LastName	DepartmentName	Email	IsActive
1	John	Doe	IT	john.doe@example.com	1
2	Jane	Smith	HR	jane.smith@example.com	1
3	Sam	Brown	Finance	sam.brown@example.com	0
4	Alice	Davis	IT	alice.davis@example.com	1
5	John	Doe	IT	john.doe@example.com	1
6	Jane	Smith	HR	jane.smith@example.com	1
7	Sam	Brown	Finance	sam.brown@example.com	0
8	Alice	Davis	IT	alice.davis@example.com	1

FirstName	LastName	Activity	LogTime
Sam	Brown	Login from IP 192.168.1.3	2025-03-12 12:00:00.000
Sam	Brown	Login from IP 192.168.1.3	2025-03-12 12:00:00.000
John	Doe	File Accessed - sensitive data	2025-03-11 10:15:00.000
John	Doe	File Accessed - sensitive data	2025-03-11 10:15:00.000
Jane	Smith	Login from IP 192.168.1.2	2025-03-10 09:00:00.000
John	Doe	Login from IP 192.168.1.1	2025-03-10 08:30:00.000
John	Doe	Login from IP 192.168.1.1	2025-03-10 08:30:00.000

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8... | CybersecurityDB | 00:00:00 | 16 rows

Item(s) Saved | Ln 79 | Col 3 | Ch 3 | INS

Connect - laptop-ekoigt17 (SQL Server 16.0.100)

- Databases
 - System Databases
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 - Database Diagrams
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 - Graph Tables
 - dbo.Departments
 - Columns
 - DepartmentID (PK, I)
 - DepartmentName (I)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.Employees
 - dbo.Logs
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 - Query Store
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```

--2. Get employees with activity logs:
SELECT e.FirstName, e.LastName, l.Activity, l.LogTime
FROM Employees e
JOIN Logs l ON e.EmployeeID = l.EmployeeID
ORDER BY l.LogTime DESC;
GO

--3. Get all active employees:
SELECT e.FirstName, e.LastName, d.DepartmentName
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID
WHERE e.IsActive = 1;
GO
  
```

100 %

Results Messages

FirstName	LastName	Activity	LogTime
Sam	Brown	Login from IP 192.168.1.3	2025-03-12 12:00:00.000
Sam	Brown	Login from IP 192.168.1.3	2025-03-12 12:00:00.000
John	Doe	File Accessed - sensitive data	2025-03-11 10:15:00.000
John	Doe	File Accessed - sensitive data	2025-03-11 10:15:00.000
Jane	Smith	Login from IP 192.168.1.2	2025-03-10 09:00:00.000
Jane	Smith	Login from IP 192.168.1.2	2025-03-10 09:00:00.000
John	Doe	Login from IP 192.168.1.1	2025-03-10 08:30:00.000
John	Doe	Login from IP 192.168.1.1	2025-03-10 08:30:00.000

FirstName	LastName	DepartmentName
John	Doe	IT
Jane	Smith	HR
Alice	Davis	IT
John	Doe	IT
Jane	Smith	HR
Alice	Davis	IT

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8... | CybersecurityDB | 00:00:00 | 14 rows

Item(s) Saved | Ln 87 | Col 3 | Ch 3 | INS

Views run below

Connect ▼ * * * * * SQL Server 16.0.100.0

- Databases
 - System Databases
 - Database Snapshots
 - CybersecurityDB
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 - Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Departments
 - Columns
 - DepartmentID (PK, I)
 - DepartmentName (I)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.Employees
 - dbo.Logs
 - Dropped Ledger Tables
 - Views
 - System Views
 - dbo.ActiveEmployees
 - dbo.RecentLogins
 - Dropped Ledger Views
 - External Resources
 - Synonyms
 - Programmability
 - Query Store
 - Service Broker
 - Storage
 - Security
 - Security
 - Server Objects

```

--Step 5: Views
--Now let's create views for common queries.
--1. Active Employees View:
CREATE VIEW ActiveEmployees AS
SELECT e.EmployeeID, e.FirstName, e.LastName, d.DepartmentName
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID
WHERE e.IsActive = 1;
GO

--2. Recent Logins View:
CREATE VIEW RecentLogins AS
SELECT e.FirstName, e.LastName, l.Activity, l.LogTime
FROM Employees e
JOIN Logs l ON e.EmployeeID = l.EmployeeID
WHERE l.Activity LIKE 'Login%';
GO

--Running above views
SELECT * FROM ActiveEmployees;

SELECT * FROM RecentLogins
ORDER BY LogTime DESC;

```

100 %

Results Messages

EmployeeID	FirstName	LastName	DepartmentName
1	John	Doe	IT
2	Jane	Smith	HR
3	Alice	Davis	IT
4	John	Doe	IT
5	Jane	Smith	HR
6	Alice	Davis	IT

FirstName	LastName	Activity	LogTime
Sam	Brown	Login from IP 192.168.1.3	2025-03-12 12:00:00.000
Sam	Brown	Login from IP 192.168.1.3	2025-03-12 12:00:00.000
Jane	Smith	Login from IP 192.168.1.2	2025-03-10 09:00:00.000
Jane	Smith	Login from IP 192.168.1.2	2025-03-10 09:00:00.000
John	Doe	Login from IP 192.168.1.1	2025-03-10 08:30:00.000
John	Doe	Login from IP 192.168.1.1	2025-03-10 08:30:00.000

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8 ... | CybersecurityDB | 00:00:00 | 12 rows

Ready Ln 111 Col 1 Ch 1 INS

CTEs Run Below

Connect ▼ * * * * * SQL Server 16.0.1000.6

- Databases
 - System Databases
 - Database Snapshots
 - CybersecurityDB
 - Database Diagrams
 - Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Departments
 - dbo.Employees
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 - Programmability
 - Query Store
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 - Security
 - Security
 - Server Objects
 - Replication
 - Always On High Availability
 - Management
 - Integration Services Catalogs
 - SQL Server Agent (Agent XPs disabled)
 - XEvent Profiler

```

--Step 6: Common Table Expressions (CTEs)
--CTEs are useful for complex queries. Let's create some CTE examples.
--1. CTE to Find Employees with Recent Activity (e.g., last 24 hours):
WITH RecentActivity AS (
    SELECT e.EmployeeID, e.FirstName, e.LastName, l.Activity, l.LogTime
    FROM Employees e
    JOIN Logs l ON e.EmployeeID = l.EmployeeID
    WHERE l.LogTime > DATEADD(DAY, -1, GETDATE())
)
SELECT * FROM RecentActivity;
GO

--2. CTE to Get Employees with Multiple Logins:
WITH MultipleLogins AS (
    SELECT e.EmployeeID, e.FirstName, e.LastName, COUNT(l.LogID) AS LoginCount
    FROM Employees e
    JOIN Logs l ON e.EmployeeID = l.EmployeeID
    WHERE l.Activity LIKE 'Login%'
    GROUP BY e.EmployeeID, e.FirstName, e.LastName
    HAVING COUNT(l.LogID) > 1
)

```

100 %

Results Messages

EmployeeID	FirstName	LastName	LoginCount
1	John	Doe	2
2	Jane	Smith	2
3	Sam	Brown	2

Query executed successfully.

laptop-ekoigt17 (16.0 RTM) | LAPTOP-EKOIGT17\pine8 ... | CybersecurityDB | 00:00:00 | 3 rows

Ready Ln 136 Col 3 Ch 3 INS